

Notes: Greenwood, Hanson, and Stein (JF, 2010)

A Gap-Filling Theory of Corporate Debt Maturity Choice

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1 Big picture

- **Question.** Why does the maturity of corporate debt vary so much over time? The paper asks whether firms choose debt maturity partly in response to macro-level supply shocks in the government bond market.
- **Core idea.** Firms act as macro liquidity providers. When the government issues relatively more long-term debt, firms fill the supply gap by issuing relatively more short-term debt. When the government issues relatively more short-term debt, firms fill the gap by issuing more long-term debt.
- **Mechanism.** The bond market is partly segmented because some investors have preferred habitats at particular maturities. Arbitrage capital is limited, so large maturity-supply shocks from the Treasury affect expected returns across the yield curve. Corporate issuers, who already need to borrow, can absorb some of those shocks by changing the maturity of their debt.
- **Key comparative advantage.** The theory does not require corporate managers to be better forecasters than bond investors. Their advantage is risk-absorption capacity: they can adjust the maturity of debt they issue, especially when deviating from their target maturity is not very costly.
- **Main prediction 1: gap filling.** Corporate debt maturity and government debt maturity should move in opposite directions. More long-term government debt should be associated with less long-term corporate debt issuance.
- **Main prediction 2: time variation.** Gap filling should be stronger when the government debt market is large relative to the overall debt market. A larger government supply shock creates a larger reward for liquidity provision.
- **Main prediction 3: cross-section.** Firms with stronger balance sheets and more financial flexibility should gap-fill more aggressively because the cost of deviating from their preferred maturity structure is lower.
- **Main prediction 4: market timing.** Corporate debt maturity may forecast bond returns, but not because managers have special forecasting skill. It forecasts returns because it mirrors government maturity-supply shocks, which are the primitive source of return predictability in the model.
- **Empirical setting.** The main sample is annual U.S. data from 1963 to 2005. The authors combine Flow of Funds corporate debt data, Compustat firm-level data, and CRSP Treasury data. Return-predictability tests use a longer 1953 to 2005 sample.
- **Economic takeaway.** Debt market timing can be understood as liquidity provision. The corporate sector absorbs part of the duration imbalance created by Treasury financing decisions, and this behavior is strongest when supply shocks are large and when firms are financially flexible.

2 Incremental contribution of the paper

- **From market timing to liquidity provision.** Earlier work, especially Baker, Greenwood, and Wurgler (2003), shows that corporate debt maturity predicts bond returns. This paper

explains why that can happen without assuming that corporate managers have superior information or forecasting ability.

- **A corporate-finance extension of preferred-habitat asset pricing.** The paper adds an active corporate issuing sector to preferred-habitat and limited-arbitrage models of the term structure. Government maturity supply affects expected returns, and firms respond by changing their own maturity supply.
- **A new theory of aggregate debt maturity choice.** Traditional maturity theories emphasize firm-level determinants such as asset-liability matching, signaling, liquidation risk, or financial constraints. This paper focuses on aggregate time-series variation and shows how corporate maturity choice can be driven by government debt management.
- **An empirical supply-shock test.** The paper identifies government debt maturity as a large observable supply shock and tests whether firms move in the opposite direction. The estimated effect implies that firms fill roughly 30% to 40% of the gap created by changes in government debt maturity.
- **A scale test that distinguishes the mechanism.** If the relation were only informational, the raw size of the government bond market would not matter. The paper shows that gap filling is stronger when government debt is larger relative to GDP or total credit-market debt, supporting the supply-shock interpretation.
- **A cross-sectional test of financial flexibility.** The paper uses Compustat to show that larger firms, dividend payers, high-cash-flow firms, high-cash firms, and low- Q firms generally gap-fill more strongly. This is consistent with lower costs of deviating from maturity targets.
- **A reinterpretation of corporate return predictability.** Corporate maturity loses predictive power for bond returns once government maturity is included. This supports the view that corporate maturity is a mirror of government supply shocks rather than an independent source of forecasting ability.
- **A broader implication for corporate finance.** The paper reframes firms as possible macro arbitrageurs or liquidity providers. This connects capital-structure decisions to asset-pricing frictions and to the limits of arbitrage.

3 Model and economic mechanism

3.1 Preferred habitat and limited arbitrage

The model has short-term and long-term default-free bonds. Preferred-habitat investors have an inelastic demand for long-term bonds, while the government supplies long-term bonds. Let

$$g = G - L,$$

where G is the government supply of long-term bonds and L is preferred-habitat demand. A positive g means excess long-term government supply.

Arbitrageurs can borrow short and buy long, or borrow long and buy short, but they are risk averse and have limited risk-bearing capacity. Their demand for long bonds rises when long bonds offer

a higher expected return premium, but because arbitrage is limited, supply shocks are not fully absorbed.

The central asset-pricing implication is:

- when the supply of long-term government debt rises relative to preferred-habitat demand, long-term bonds must offer a higher expected return;
- when the supply of short-term government debt rises, short-term instruments become relatively abundant and firms have an incentive to issue more long-term debt.

Interpretation: government debt management changes the relative supply of duration. Because arbitrage capital is limited, these supply shifts affect expected returns and borrowing costs.

3.2 Corporate issuers

Firms need to raise a fixed amount of debt C . They choose a fraction f of this debt to issue long term and $1 - f$ to issue short term. Their target maturity share is z . Deviating from the target creates costs,

$$\frac{\theta C}{2}(f - z)^2,$$

where θ measures the cost of deviating from the target. A low θ firm is financially flexible; a high θ firm is more constrained and has a stronger reason to stay close to its target maturity structure.

The corporate choice trades off expected borrowing costs against the cost of moving away from the target. When long-term debt is expensive relative to short-term debt, firms reduce f and issue less long-term debt. When long-term debt is cheap, firms increase f .

A useful limiting case is the Modigliani-Miller benchmark. If θ is close to zero, firms are almost indifferent about maturity around their target. Then even small expected-return differences induce a large maturity response. Firms can absorb government supply shocks very elastically.

3.3 Four propositions

- **Proposition 1: gap filling.** $\partial f^*/\partial g < 0$. When the government issues more long-term debt, firms tilt away from long-term debt.
- **Proposition 2: time variation in gap filling.** Gap filling is stronger when the government debt market is larger relative to the corporate debt market. In dollars, firms must change their own maturity mix more when government supply shocks are larger.
- **Proposition 3: cross-section of gap filling.** Firms with stronger balance sheets should respond more elastically to government maturity supply because their cost of deviating from target maturity is lower.
- **Proposition 4: origin of market timing ability.** Corporate maturity forecasts bond returns because it responds to government supply shocks. Once government debt maturity is included in the return-forecasting regression, the incremental predictive power of corporate maturity should fall.

4 Data and measurement

4.1 Corporate debt maturity from Flow of Funds

The Flow of Funds data cover the nonfarm, nonfinancial corporate business sector. The authors define short-term corporate debt as commercial paper, bank loans not elsewhere classified, and other loans and advances. These instruments generally have maturities below one year.

Long-term corporate debt is defined as industrial revenue bonds, corporate bonds, and mortgages. The first Flow of Funds maturity measure is the long-term corporate *level share*,

$$\frac{D_{L,t}^C}{D_t^C}.$$

The second Flow of Funds measure is the long-term corporate *issue share*. Long-term issues are approximated as

$$d_{L,t}^C = (D_{L,t}^C - D_{L,t-1}^C) + 0.1D_{L,t-1}^C,$$

which assumes an average long-term debt maturity of 10 years. The issue share is

$$\frac{d_{L,t}^C}{d_t^C}.$$

Interpretation: the level share measures the stock of outstanding long-term debt. The issue share is closer to the marginal maturity choice, which is the natural object in a dynamic setting with adjustment costs.

4.2 Corporate debt maturity from Compustat

Compustat provides firm-level data for public companies. The authors aggregate across nonfinancial firms and define long-term debt to include long-term borrowings plus the portion of long-term debt about to mature. This produces a Compustat long-term level share comparable to the Flow of Funds level measure.

The key advantage of Compustat is disaggregation. It allows the authors to construct maturity shares for groups of firms sorted by size, dividends, cash flow, cash balances, Tobin's Q , and leverage. This is essential for testing the cross-sectional prediction that financially flexible firms gap-fill more.

Interpretation: Flow of Funds is better for aggregate coverage, while Compustat is better for cross-sectional heterogeneity.

4.3 Government debt maturity from CRSP

The government debt maturity data come from the CRSP U.S. Treasury Database. The authors decompose each Treasury security into principal and coupon payments and aggregate payments due at different horizons.

The main government maturity measure is the long-term government share,

$$\frac{D_{L,t}^G}{D_t^G},$$

where the numerator is the value of principal and coupon payments due in more than one year. The paper also uses M , the face-value-weighted average maturity of principal payments.

Interpretation: the government maturity variable is meant to proxy for the relative supply of long-term government debt, which is the key primitive shock in the model.

4.4 Interest rates, returns, and controls

The main controls include the one-year Treasury yield $y_{S,t}$, the term spread $y_{L,t} - y_{S,t}$, the Moody's Baa credit spread, GDP growth, recession indicators, government investment, and corporate investment.

For the return-predictability tests, the dependent variables are future excess returns on long-term Treasury bonds over one-, two-, and three-year horizons.

Interpretation: the controls help distinguish gap filling from simpler stories in which firms merely react to the level of interest rates, the slope of the yield curve, credit conditions, or the business cycle.

5 Models, empirical specifications, and identification

5.1 Baseline gap-filling regressions

The baseline aggregate specification can be written as

$$\text{Corporate maturity}_t = \alpha + \beta \text{Government maturity}_t + \Gamma' X_t + \varepsilon_t.$$

The prediction is $\beta < 0$. The dependent variable is alternately the Flow of Funds corporate long-term level share, the Flow of Funds long-term issue share, or the Compustat long-term level share. Government maturity is measured either by $D_{L,t}^G/D_t^G$ or by the weighted average maturity M .

The richer specification includes the short-term Treasury yield, the term spread, and a linear time trend. Standard errors are Newey-West with two years of lags.

Interpretation: a negative coefficient says that firms issue or hold less long-term debt when the government supplies more long-term debt.

5.2 Robustness and endogeneity

The key identification concern is that government maturity may be endogenous. For example, the Treasury and corporations might both respond to omitted macroeconomic forces. The paper addresses this concern in several ways:

- adding interest-rate, credit-spread, business-cycle, GDP-growth, and investment controls;

- using alternative corporate maturity measures and alternative government maturity measures;
- checking subsamples and a longer Flow of Funds sample;
- adjusting for mortgages and interest-rate swaps;
- examining financial firms separately;
- estimating differenced and Prais-Winsten GLS specifications;
- using the government debt-to-GDP ratio as an instrument for government debt maturity.

The IV logic is that when government financing needs are larger, the Treasury tends to extend maturity to manage rollover risk. If past fiscal policy affects corporate maturity only through government maturity supply, debt-to-GDP is a valid instrument.

Interpretation: the empirical design is not a randomized experiment, but the robustness and IV evidence make it harder to explain the results purely by omitted demand or macro conditions.

5.3 Time-series, cross-sectional, and return-predictability tests

To test time variation in gap filling, the paper estimates regressions with an interaction between government maturity and the size of the government bond market:

$$\frac{d_{L,t}^C}{d_t^C} = \alpha + b \frac{D_{L,t}^G}{D_t^G} + c \text{Scale}_t + d \left(\text{Scale}_t \times \frac{D_{L,t}^G}{D_t^G} \right) + e \text{time} + f \left(\text{time} \times \frac{D_{L,t}^G}{D_t^G} \right) + \theta' X_t + \varepsilon_t.$$

The prediction is $d < 0$: gap filling is stronger when the scale of government debt is larger.

For the cross-section, the paper constructs Compustat maturity shares for financially flexible and less flexible firms. Financial flexibility is proxied by market capitalization, dividend payer status, cash flow/assets, cash/assets, Tobin's Q , and leverage.

For return predictability, the paper estimates regressions of future excess bond returns on government maturity, corporate maturity, or both. Proposition 4 predicts that the corporate maturity coefficient should shrink when government maturity is added.

6 Tables and figures

6.1 Table I: Summary Statistics

Read:

- Table I summarizes the main corporate maturity, government maturity, interest-rate, return, and macro control variables from 1963 to 2005.
- The Flow of Funds long-term corporate level share averages 61.51%, while the Flow of Funds long-term issue share averages 21.57%.
- The Compustat long-term corporate level share is higher, with a mean of 83.41%, consistent with public firms having better access to long-term finance.

- The government long-term share averages 59.09%, and the weighted average maturity of government bonds averages 4.51 years.
- Government maturity variables are highly persistent: the long-term government share has autocorrelation 0.95, and weighted average maturity has autocorrelation 0.96.
- The Flow of Funds issue share is less persistent than the level share, which motivates using issuance as a marginal maturity-choice measure.

Economic message: Table I establishes the objects of interest and the econometric challenge. The maturity series contain meaningful variation, but some are very persistent, so the paper uses multiple measures, Newey-West standard errors, differenced specifications, and GLS checks.

Table I
Summary Statistics

This table presents means, medians, standard deviations, extreme values, and time-series autocorrelations (denoted ρ) of variables between 1963 and 2005. Panel A shows the corporate long-term level share, and the corporate long-term issue share, based on *Flow of Funds* (FOF) data. All FOF short-term debt is assumed to be new short-term issues. FOF long-term issues are defined as the change in FOF long-term debt plus one-tenth of lagged FOF long-term debt. Panel B shows the corresponding levels measure from Compustat. Compustat debt is the sum of long-term debt and debt in current liabilities. Long-term debt is the sum of all long-term borrowings, plus debt that was originally issued long term but that is about to retire. Panel C summarizes measures of public debt maturity, estimated using the CRSP government bond database. D_L^G/D^G is the fraction of principal and coupon payments that is due in more than 1 year. M is the face value-weighted maturity of government bonds. Panel D summarizes interest rate conditions: y_{St} is the log yield on 1-year Treasuries, $y_{Lt} - y_{St}$ is the spread between the log yields of the 20-year Treasury bond and the 1-year Treasury bond, $R_{Lt+1} - y_{St}$ is the log 1-year forward excess bond return, and *Credit spread* is the Moody's Baa yield minus the average yield on long-term Treasuries. Panel E summarizes the ratio of government debt to GDP; the ratio of government debt to total credit market debt; annual GDP growth; a recession dummy based on NBER dating conventions; the ratio of government investment to GDP; and net private investment, calculated as the percentage change in net private property, plant, and equipment. The last two variables are calculated using data from the Bureau of Economic Analysis. All variables, except for M and the *Recession dummy*, are expressed in percentage terms.

	Mean	Median	SD	Min	Max	ρ
Panel A: FOF Corporate Debt Maturity %						
Levels: D_L^C/D^C	61.51	60.93	4.97	53.46	73.12	0.85
Issues: d_L^C/d^C	21.57	21.28	4.14	14.75	30.13	0.58
Panel B: Compustat Corporate Debt Maturity %						
Levels: D_L^C/D^C	83.41	83.69	3.36	77.00	89.75	0.76
Panel C: Gov. Debt Maturity %						
D_L^G/D^G	59.09	58.78	8.94	41.74	72.48	0.95
M (years)	4.51	4.57	0.90	2.82	5.75	0.96
Panel D: Short Rate, Term Spread, Subsequent Bond Returns, and Credit Spread (%)						
y_{St}	6.01	5.41	2.99	0.96	16.86	0.74
$y_{Lt} - y_{St}$	0.87	0.73	1.41	-1.60	3.75	0.63
$R_{Lt+1} - y_{St}$	0.98	0.22	9.81	-15.21	21.01	-0.10
<i>Credit spread</i>	2.01	1.84	0.80	0.59	3.85	0.69
Panel E: Other Controls (%)						
D^G/GDP	34.63	34.08	7.73	22.46	48.67	0.96
D^G/D	19.62	19.73	3.11	15.20	26.96	0.85
$\Delta \text{Log}(GDP)$	3.25	3.48	1.99	-1.95	6.94	0.25
<i>Recession dummy</i>	0.26	0.00	0.44	0.00	1.00	0.38
<i>GovInv/GDP</i>	1.42	1.27	0.47	0.89	2.60	0.86
<i>CorpInv/PPE</i> _{t-1}	7.29	6.30	3.49	2.13	17.80	0.76

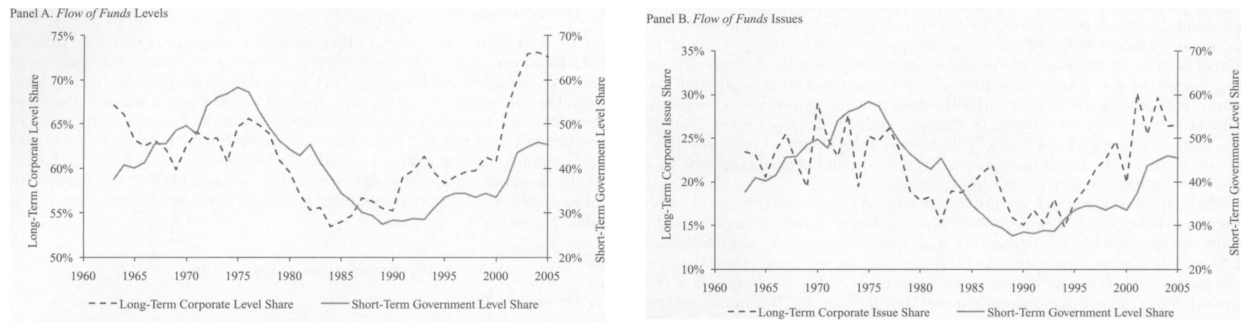
Table I: Summary Statistics.

6.2 Figure 1: Corporate and government debt maturity

Read:

- Figure 1 provides the first visual evidence for gap filling.
- The dashed line is a corporate long-term debt share; the solid line is the share of government debt with maturity of one year or less.
- Because the government variable is inverted into a short-term share, positive co-movement in the figure corresponds to a negative relation between corporate and government long-term maturity.
- Panel A uses Flow of Funds corporate levels, Panel B uses Flow of Funds corporate issues, and Panel C uses Compustat corporate levels.
- The co-movement is visible in all three panels: when the government tilts toward shorter maturity, firms tend to tilt toward longer maturity, and vice versa.

Economic message: The figure illustrates the central gap-filling pattern before any regression controls. Corporate maturity choices appear to offset government maturity supply.



(a) Figure 1, Panel A: Flow of Funds levels

(b) Figure 1, Panel B: Flow of Funds issues

Figure 1, Panels A and B: corporate maturity and short-term government share.

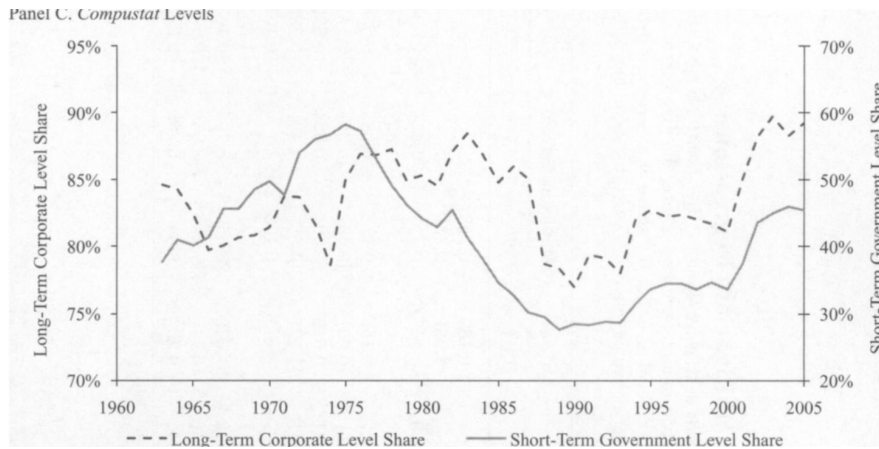


Figure 1, Panel C: Compustat levels.

6.3 Table II: The maturity of corporate and government debt, univariate regressions

Read:

- Table II regresses each corporate maturity measure on government maturity, without additional controls.
- All six coefficients are negative, as predicted by the gap-filling theory.
- Using the government long-term share, the coefficients are -0.262 for Flow of Funds levels, -0.249 for Flow of Funds issues, and -0.147 for Compustat levels.
- The Flow of Funds results are statistically strong, with t -statistics of -3.64 and -4.21 .
- Using weighted average maturity M produces the same qualitative pattern.
- The paper interprets the magnitudes as implying that firms fill roughly 27% of the maturity gap created by variation in government debt maturity in the simple univariate setting.

Economic message: Table II is the first formal test of Proposition 1. It shows that corporate maturity moves opposite to government maturity, consistent with firms supplying duration when the government removes it and removing duration when the government supplies it.

TABLE II
The Maturity of Corporate and Government Debt, 1963–2005:
Univariate Regressions

This table presents OLS regressions of the maturity of corporate debt on the maturity of government debt. The dependent variable is alternately the *Flow of Funds* (FOF) corporate long-term level share, the FOF corporate long-term issue share, or the Compustat corporate long-term level share. The maturity of government debt is defined as either the share of government debt and coupon payments with maturity of 1 year or more (D_L^G/D^G), or the dollar-weighted maturity of principal payments (M). t -statistics, in brackets, are based on Newey-West (1987) standard errors allowing for 2 years of lags.

	FOF: Levels		FOF: Issues		Compustat: Levels	
	(1)	(2)	(3)	(4)	(5)	(6)
D_L^G/D^G	-0.262 [-3.64]		-0.249 [-4.21]		-0.147 [-1.83]	
M		-1.804 [-2.64]		-1.949 [-2.85]		-1.272 [-1.67]
<i>Constant</i>	76.971 [16.58]	69.656 [22.08]	36.286 [10.16]	30.372 [10.42]	92.090 [18.55]	89.148 [24.62]
R^2	0.22	0.11	0.29	0.18	0.15	0.12

Table II: Univariate regressions.

6.4 Table III: Multivariate regressions

Read:

- Table III adds the short-term Treasury yield, the term spread, and in some specifications a linear time trend.
- The coefficient on government maturity remains negative across all corporate maturity measures.
- For Flow of Funds levels, the coefficient on the government long-term share is -0.296 with interest-rate controls and -0.387 with interest-rate controls plus trend.

- For Flow of Funds issues, the corresponding coefficients are -0.278 and -0.318 .
- For Compustat levels, the coefficients are -0.169 and -0.228 .
- Adding controls generally improves statistical significance, suggesting that the gap-filling relation is not driven by the level of rates, the slope of the yield curve, or a simple time trend.

Economic message: Table III strengthens the core result. Corporate maturity is negatively related to government maturity even after controlling for interest-rate conditions that could otherwise affect firms' borrowing choices.

Table III
The Maturity of Corporate and Government Debts, 1963–2005: Multivariate Regressions

This table presents OLS regressions of the maturity of corporate debt on the maturity of government debt, controlling for the short-term rate, the term spread, and a time trend. The dependent variable is alternately the *Flow of Funds* (FOF) corporate long-term level share, the FOF corporate long-term issue share, or the Compustat long-term level share. The maturity of government debt is defined as either the share of government debt and coupon payments with maturity of 1 year or more (D_L^G/D^G), or the dollar-weighted maturity of principal payments (M). t -statistics, in brackets, are based on Newey-West (1987) standard errors allowing for 2 years of lags.

	FOF: Levels				FOF: Issues				Compustat: Levels			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
D_L^G/D^G	-0.296 [-5.14]	-0.387 [-5.45]			-0.278 [-5.00]	-0.318 [-5.77]			-0.169 [-1.96]	-0.228 [-2.33]		
M			-2.540 [-4.31]	-3.488 [-4.03]			-2.526 [-4.68]	-2.939 [-5.64]			-1.474 [-1.80]	-2.094 [-2.05]
y_{St}	-1.214 [-2.93]	-1.263 [-3.55]	-1.317 [-2.87]	-1.404 [-3.43]	-0.815 [-5.02]	-0.836 [-5.81]	-0.919 [-4.97]	-0.957 [-5.45]	0.155 [0.60]	0.123 [0.48]	0.095 [0.35]	0.038 [0.14]
$y_{Lt} - y_{St}$	-0.613 [-1.11]	-1.257 [-2.72]	-0.781 [-1.30]	-1.436 [-2.94]	-0.207 [-0.48]	-0.486 [-1.08]	-0.355 [-0.74]	-0.641 [-1.31]	0.919 [1.77]	0.504 [0.88]	0.824 [1.48]	0.395 [0.66]
$Trend$		0.160 [2.26]		0.154 [1.78]		0.069 [2.07]		0.067 [1.71]		0.103 [1.62]		0.101 [1.38]
$Constant$	86.825 [17.14]	89.736 [22.70]	81.581 [16.76]	83.723 [21.30]	43.109 [13.74]	44.369 [14.83]	38.807 [15.25]	39.742 [17.11]	91.679 [16.14]	93.550 [17.26]	88.765 [18.93]	90.168 [20.60]
R^2	0.63	0.73	0.55	0.64	0.59	0.61	0.52	0.54	0.25	0.34	0.20	0.29

Table III: Multivariate regressions.

6.5 Table IV: Robustness checks

Read:

- Table IV varies the baseline specification in many ways while reporting the slope on the long-term government share.
- The baseline estimates are -0.387 for Flow of Funds levels, -0.318 for Flow of Funds issues, and -0.228 for Compustat levels.
- The results survive controls for business-cycle leads and lags, GDP growth leads and lags, credit spreads, government investment, and corporate investment.
- Results are also robust to adding marketable mortgages to the government supply measure and to adjusting corporate maturity for net swap activity.
- Gap filling is stronger in the second half of the sample, especially for the Compustat measure.
- The IV estimates using government debt-to-GDP as an instrument are close to the OLS estimates. For example, the IV coefficient for Flow of Funds levels is -0.395 , compared with the baseline OLS value of -0.387 .

Economic message: Table IV addresses the main alternative explanations. The negative relation is not fragile, and it remains when the authors use an instrument designed to isolate government debt maturity supply.

Robustness Checks

This table presents regressions of the maturity of corporate debt on the maturity of government debt. We vary the basic specification in row (1) by: (2) using only the first half of the sample period; (3) using only the second half of the sample period; (4) extending the *Flow of Funds* (FOF) data back to 1953; (5) controlling for two leads and two lags of the NBER recession dummy; (6) controlling for two leads and two lags of changes in log GDP; (7) controlling for the credit spread, defined as the Moody's Baa yield minus the average yield on long-term Treasuries; (8) controlling for government investment scaled by total government fixed capital; (9) controlling for aggregate corporate investment; (10) adjusting our corporate maturity measure for marketable mortgages, which we count as long-term debt; (11) adjusting our corporate maturity measure for net swap activity; and (12) using Compustat financial firms only. In the last two rows, we instrument for government debt maturity using the ratio of government debt to GDP where (13) presents instrumental variable estimates for our baseline specification, and (14) adds the credit spread as a control. For each regression, we report the slope coefficient on the long-term government share, b , and its associated t -statistic. All regressions include a constant term and controls for the short-term rate, the term spread, and a time trend. t -statistics are based on Newey-West (1987) standard errors allowing for 2 years of lags.

	FOF: Levels			FOF: Issues			Compustat: Levels		
	b	$[t]$	R^2	b	$[t]$	R^2	b	$[t]$	R^2
OLS:									
(1) Baseline	-0.387	[-5.45]	0.73	-0.318	[-5.77]	0.61	-0.228	[-2.33]	0.34
(2) First half (1963–1983)	-0.203	[-1.84]	0.58	-0.266	[-3.34]	0.52	0.163	[1.88]	0.58
(3) Second half (1984–2005)	-0.371	[-3.63]	0.95	-0.477	[-2.97]	0.71	-0.787	[-11.68]	0.92
(4) Long sample (1953–2005)	-0.303	[-3.99]	0.70	-0.269	[-4.72]	0.60	–	–	–
(5) Business cycle leads and lags	-0.383	[-4.64]	0.74	-0.381	[-8.13]	0.77	-0.305	[-4.02]	0.62
(6) $\Delta \text{Log}(\text{GDP})$ leads and lags	-0.373	[-5.20]	0.74	-0.383	[-6.30]	0.73	-0.280	[-3.37]	0.57
(7) Control for credit spread	-0.466	[-7.52]	0.76	-0.328	[-5.80]	0.61	-0.308	[-3.34]	0.41
(8) Control for Gov. investment	-0.532	[-5.12]	0.76	-0.418	[-5.11]	0.64	-0.312	[-3.23]	0.37
(9) Control for corp. investment	-0.367	[-4.37]	0.73	-0.338	[-4.41]	0.61	-0.332	[-2.58]	0.38
(10) Adjust for mortgages	-0.509	[-4.00]	0.61	-0.434	[-5.00]	0.52	-0.293	[-1.87]	0.25
(11) Adjust for swaps	-0.309	[-5.83]	0.73	–	–	–	-0.149	[-1.70]	0.21
(12) Compustat financial firms	–	–	–	–	–	–	-0.298	[-2.06]	0.63
Instrumental Variables:									
(13) Baseline	-0.395	[-4.89]	0.73	-0.402	[-5.33]	0.59	-0.242	[-2.59]	0.34
(14) Control for credit spread	-0.539	[-6.41]	0.75	-0.488	[-4.87]	0.55	-0.392	[-4.35]	0.39

Table IV: Robustness checks.

6.6 Tables V and VI: Differenced and GLS regressions

Read:

- Table V estimates differenced specifications over windows from one to five years.
- The one-year differenced estimates are weaker, but the relation becomes stronger as the window expands, suggesting gradual adjustment rather than an entirely contemporaneous response.
- For the Flow of Funds issue share, the coefficient is -0.289 with a five-year window and a t -statistic of -5.10 .
- For Flow of Funds levels, the five-year differenced coefficient is -0.325 with a t -statistic of -1.94 .
- Table VI uses Prais-Winsten GLS. The Flow of Funds issue-share results are very similar to OLS: with full controls the coefficient is -0.316 and the t -statistic is -6.16 .
- The level-share GLS specifications are less informative because residual persistence is extremely high, making GLS close to first-differencing.

Economic message: Tables V and VI show that the main result is not merely a low-frequency visual coincidence. Issuance responds to government maturity changes, but the adjustment appears to occur over several years.

Table V
Differenced Regressions

This table presents regressions of the following form:

$$d_{L,t}^C/d_t^C = a + b \cdot \Delta_k(D_{L,t}^C/D_t^C) + \varepsilon_t$$

$$\Delta_k(D_{L,t}^C/D_t^C) = a + b \cdot \Delta_k(D_{L,t}^C/D_t^C) + \varepsilon_t.$$

The dependent variable is alternately $d_{L,t}^C/d_t^C$, the *Flow of Funds* (FOF) long-term corporate issue share, or $\Delta_k(D_{L,t}^C/D_t^C)$, the change in either the FOF or the Compustat long-term corporate level share over a k -year window. The independent variable is $\Delta_k(D_{L,t}^C/D_t^C)$, the change in the long-term government share over a k -year window. t -statistics are based on Newey-West (1987) standard errors allowing for k years of lags in the regressions on $\Delta_k(D_{L,t}^C/D_t^C)$.

	FOF Issues			Changes in FOF Levels			Changes in Compustat Levels		
	b	$[t]$	R^2	b	$[t]$	R^2	b	$[t]$	R^2
$k = 1$ lag	-0.309	[-1.33]	0.04	-0.179	[-1.30]	0.06	-0.211	[-1.84]	0.07
$k = 2$ lags	-0.331	[-2.26]	0.12	-0.265	[-1.64]	0.13	-0.273	[-2.10]	0.13
$k = 3$ lags	-0.287	[-2.65]	0.16	-0.282	[-1.63]	0.16	-0.237	[-1.78]	0.13
$k = 4$ lags	-0.285	[-3.92]	0.25	-0.308	[-1.88]	0.21	-0.228	[-1.78]	0.16
$k = 5$ lags	-0.289	[-5.10]	0.33	-0.325	[-1.94]	0.24	-0.230	[-1.76]	0.19

(a) Table V: Differenced regressions

Table VI
GLS Regressions

This table presents Prais-Winsten GLS regressions of the maturity of corporate debt on the maturity of government debt, controlling for the short-term rate, the term spread, and a time trend. The Prais-Winsten GLS procedure is based on the assumption that regression residuals follow an AR(1). The dependent variable is alternately the *Flow of Funds* (FOF) corporate long-term level share, the FOF corporate long-term issue share, or the Compustat long-term level share. The maturity of government debt is defined as the share of government debt and coupon payments with maturity of 1 year or more (D_t^G/D_t^C). The t -statistics, in brackets, are computed using Prais-Winsten GLS standard errors that are robust to heteroskedasticity (equivalent to robust OLS standard errors for the transformed regression). We also report the estimated first-order autocorrelation of the residuals, denoted by ρ .

	FOF: Levels			FOF: Issues			Compustat: Levels		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
D_t^C/D_t^G	-0.187 [-1.44]	-0.130 [-1.20]	-0.130 [-1.21]	-0.238 [-2.60]	-0.276 [-4.70]	-0.316 [-6.16]	-0.212 [-1.73]	-0.194 [-1.85]	-0.209 [-2.05]
ys_t		-0.290 [-1.14]	-0.300 [-1.15]		-0.780 [-3.78]	-0.826 [-4.60]		-0.274 [-1.14]	-0.259 [-1.04]
$ys_t - ys_t$		0.299 [0.79]	0.284 [0.73]		-0.100 [-0.21]	-0.437 [-1.02]		0.155 [0.40]	0.145 [0.36]
<i>Trend</i>			0.101 [0.65]			0.066 [1.95]			0.087 [1.16]
<i>Constant</i>	77.087 [7.79]	75.492 [9.72]	71.445 [9.22]	35.753 [6.18]	42.698 [11.97]	43.478 [14.06]	96.499 [12.51]	96.886 [14.32]	94.831 [14.33]
R^2	0.62	0.66	0.73	0.25	0.53	0.59	0.87	0.88	0.90
ρ	0.96	0.97	0.96	0.43	0.16	0.05	0.80	0.82	0.77

(b) Table VI: GLS regressions

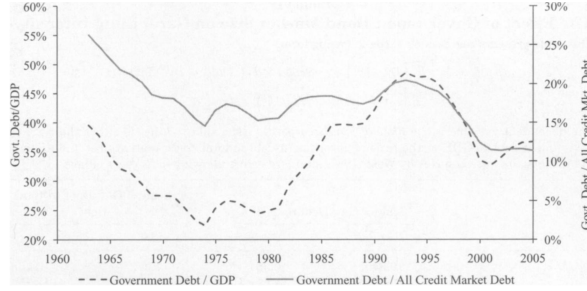
Tables V and VI: persistence, differencing, and GLS checks.

6.7 Figure 2 and Table VII: Time variation in gap-filling intensity

Read:

- Figure 2 plots two measures of government bond-market size: government debt-to-GDP and government debt-to-total-credit-market-debt.
- Table VII tests whether gap filling is more intense when the government debt market is larger.
- The key coefficient is the interaction between the scale variable and the long-term government share.
- The interaction coefficient is negative in all four specifications, as predicted by Proposition 2.
- The estimates are statistically significant in three of the four specifications and marginal in the remaining one.
- This scale result is important because it distinguishes supply-driven gap filling from a pure information story. In a pure information story, the raw size of government debt supply should not matter in the same way.

Economic message: When government debt supply is large, corporate gap filling is stronger. This supports the view that firms respond to the dollar size of maturity-supply shocks.



(a) Figure 2: Debt market size

Table VII
The Effect of Government Bond Market Size on Gap-Filling Intensity
This table presents regressions of the following form:

$$d_{L,t}^C/d_t^C = a + b \cdot (D_{L,t}^G/D_t^G) + c \cdot Scale_t + d \cdot (Scale_t \times D_{L,t}^G/D_t^G) + e \cdot time + f \cdot (time \times D_{L,t}^G/D_t^G) + \theta' \mathbf{x}_t + \varepsilon_t.$$

The dependent variable is the *Flow of Funds* corporate issue share. $Scale_t$ is either the ratio of government debt to GDP, or the ratio of government debt to total credit market debt. t -statistics, in brackets, are based on Newey-West (1987) standard errors allowing for 2 years of lags.

	Scale = Gov. Debt to GDP		Scale = Gov. Debt to Total Debt	
	(1)	(2)	(3)	(4)
$D_{L,t}^G/D_t^G$	0.640 [2.79]	0.351 [1.49]	1.272 [3.07]	0.768 [2.15]
$Scale$	2.906 [4.41]	0.957 [1.32]	5.427 [3.85]	2.495 [1.89]
$(Scale) \times (D_{L,t}^G/D_t^G)$	-4.400 [-4.49]	-1.941 [-1.77]	-8.397 [-3.83]	-4.794 [-2.37]
$Time$	-0.916 [-1.69]	-0.268 [-0.68]	0.503 [1.19]	0.200 [0.80]
$Time \times (D_{L,t}^G/D_t^G)$	0.017 [1.77]	0.006 [0.87]	-0.005 [-0.71]	-0.003 [-0.77]
y_{St}		-0.920 [-4.08]		-0.904 [-4.66]
$y_{Lt} - y_{St}$		-0.293 [-0.58]		-0.322 [-0.70]
$Constant$	-26.504 [-2.09]	12.442 [0.76]	-65.239 [-2.55]	-10.706 [0.44]
R^2	0.52	0.71	0.49	0.71

(b) Table VII: Scale interaction regressions

Figure 2 and Table VII: government debt scale and gap-filling intensity.

6.8 Table VIII and Figure 3: Cross-section of gap filling

Read:

- Table VIII disaggregates Compustat firms by proxies for financial flexibility.
- Large firms gap-fill more than small firms. The coefficient is -0.286 for large firms versus 0.024 for small firms, and the difference is statistically significant.
- Dividend payers gap-fill more than nonpayers. The coefficient is -0.263 for payers versus -0.043 for nonpayers.
- High cash-flow firms, high cash-balance firms, and low- Q firms also show more negative coefficients, although not every difference is statistically significant.
- Book leverage does not produce the predicted differential; high- and low-leverage firms have very similar coefficients.
- Figure 3 illustrates two of the splits visually: large versus small firms and dividend payers versus nonpayers.

Economic message: The cross-section broadly supports Proposition 3. Firms with stronger balance sheets or more flexibility appear more willing to adjust maturity in response to government supply shocks.

Table VIII
Disaggregated Results by Firm Type, 1963–2005

This table presents OLS regressions of the Compustat long-term level share on the long-term government share, disaggregated by firm type. Each year, nonfinancial firms are classified as low (below 30th percentile) or high (greater than 70th percentile) with respect to: market capitalization, cash flow over assets, cash balances over assets, Tobin's Q , and book leverage. Firms are also classified as either dividend payers or nonpayers. These separate maturity measures for low and high firms are regressed on the long-term government share. In the first two columns of each row, we report the slope coefficient on the long-term government share, b , and its associated t -statistic from the separate low firm and high firm regressions. In the final column, we regress the difference between the level shares for the high and low groups on the government level share. The t -statistic from this regression tests the equality of coefficients between the high and low groups. All regressions include a constant term and controls for the short-term rate, the term spread, and a time trend. t -statistics are based on Newey-West (1987) standard errors allowing for 2 years of lags.

	Low		High		High – Low	
	b	$[t]$	b	$[t]$	$b^{\text{High}} - b^{\text{Low}}$	$[t]$
All Compustat nonfinancial	-0.228	[-2.33]				
Market capitalization	0.024	[0.43]	-0.286	[-2.50]	-0.310	[-2.18]
Nonpayers ("low"); payers ("high")	-0.043	[-0.83]	-0.263	[-2.30]	-0.220	[-1.91]
Cash flow/assets	0.073	[1.35]	-0.125	[-1.42]	-0.198	[-1.94]
Cash/assets	-0.059	[-0.39]	-0.215	[-2.53]	-0.156	[-1.07]
Tobin's Q	-0.318	[-3.09]	-0.063	[-0.69]	0.255	[1.97]
Leverage	-0.375	[-3.19]	-0.367	[-2.88]	0.008	[0.06]

Table VIII: Disaggregated results by firm type.

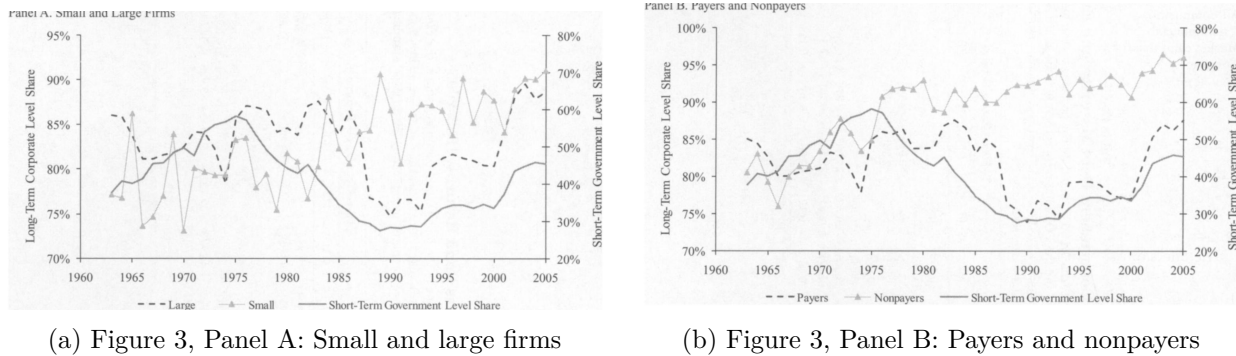


Figure 3: Long-term debt share, Compustat splits.

6.9 Table IX: Corporate debt maturity, government debt maturity, and excess bond returns

Read:

- Table IX examines whether government and corporate maturity forecast future excess returns on long-term Treasury bonds.

- Government long-term maturity positively forecasts excess bond returns. A one percentage point increase in the government long-term share predicts higher excess returns of 22.5, 52.3, and 82.4 basis points over one-, two-, and three-year horizons.
- Corporate long-term maturity predicts returns with the opposite sign, especially at two- and three-year horizons.
- The key test is the bivariate horse race. Once government maturity is included, the predictive coefficient on corporate maturity falls.
- For example, the Flow of Funds corporate issue-share coefficient at the three-year horizon falls from -1.588 to -1.045 , and its t -statistic falls from -2.74 to -1.60 .
- The same attenuation appears for the corporate level share.

Economic message: Table IX supports Proposition 4. Corporate maturity appears to forecast bond returns partly because it is a mirror image of government maturity-supply shocks, not because managers have a standalone forecasting advantage.

Table IX
Corporate Debt Maturity, Government Debt Maturity, and Excess Bond Returns, 1953–2005

Annual regressions of future log excess bond returns over 1-, 2-, and 3-year horizons on combinations of the long-term government share, the *Flow of Funds* (FOF) long-term corporate level share, and the FOF long-term corporate issue share. In the 1-year regressions, the t -statistics, in brackets, are based on heteroskedasticity-robust standard errors. In the 2- and 3-year regressions, t -statistics are computed using Newey-West (1987) standard errors allowing for autocorrelation at up to 1 and 2 years, respectively.

	One-Year Excess Returns (%)					Two-Year Excess Returns (%)					Three-Year Excess Returns (%)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
$D_{L,t}^G/D_t^G$	0.225		0.185		0.170	0.523		0.394		0.388	0.824		0.580		0.576
	[1.57]		[1.09]		[1.10]	[2.59]		[1.60]		[1.71]	[3.06]		[1.82]		[1.95]
$d_{L,t}^C/d_t^C$		-0.337	-0.163				-0.912	-0.542				-1.588	-1.045		
		[-1.00]	[-0.42]				[-2.36]	[-1.16]				[-2.74]	[-1.60]		
$D_{L,t}^C/D_t^C$				-0.312	-0.206				-0.778	-0.531				-1.408	-1.034
				[-1.41]	[-0.84]				[-2.28]	[-1.33]				[-3.08]	[-1.99]
<i>Constant</i>	-12.583	8.190	-6.630	20.280	3.565	-29.532	21.591	-9.876	50.063	11.695	-46.773	37.122	-9.173	89.926	32.510
	[-1.52]	[1.03]	[-0.40]	[1.40]	[0.17]	[-2.31]	[2.54]	[-0.42]	[2.34]	[0.34]	[-2.71]	[3.16]	[-0.30]	[3.26]	[0.73]
R^2	0.04	0.02	0.05	0.03	0.05	0.12	0.09	0.15	0.11	0.17	0.19	0.17	0.25	0.20	0.28

Table IX: Corporate debt maturity, government debt maturity, and excess bond returns.

7 Short summary

- The paper studies why aggregate corporate debt maturity changes over time.
- The theory combines preferred-habitat investors, limited arbitrage, government debt maturity supply shocks, and corporate issuers.
- The central claim is that firms fill maturity gaps created by Treasury debt management.
- When the government issues more long-term debt, firms issue less long-term debt; when the government issues more short-term debt, firms issue more long-term debt.
- Empirically, corporate and government debt maturities are strongly negatively related in annual U.S. data.
- The estimated magnitude implies that firms fill about 30% to 40% of the gap created by government maturity shocks.

- The result is robust to interest-rate controls, credit-spread controls, macro controls, investment controls, mortgage adjustments, swap adjustments, alternative samples, and IV estimation.
- Gap filling is stronger when the government debt market is large relative to GDP or total credit-market debt.
- Gap filling is stronger among financially flexible firms, especially large firms and dividend payers.
- Corporate maturity predicts bond returns partly because it reflects government maturity supply. Once government maturity is controlled for, corporate maturity’s return-predictive power weakens.
- The paper’s main contribution is to reinterpret debt market timing as macro liquidity provision rather than managerial forecasting skill.

8 Conclusion

8.1 Core conclusion

Greenwood, Hanson, and Stein show that corporate debt maturity choices respond systematically to government debt maturity. The evidence supports a gap-filling theory: firms offset maturity-supply shocks created by Treasury debt management. Corporate issuers act as macro liquidity providers in the bond market.

8.2 Economic intuition

The economic intuition is that long-term and short-term bonds are not perfect substitutes for all investors. Preferred-habitat investors demand certain maturities, and arbitrageurs have limited capital. When the Treasury changes the maturity composition of its debt, expected returns across maturities shift. Firms that already need to borrow can take the other side by changing whether they issue short-term or long-term debt.

This does not require firms to know more than investors. A firm can respond to publicly observable government supply shocks because its comparative advantage is not information, but risk absorption. The more flexible the firm, the easier it is to deviate from target debt maturity and provide that liquidity.

8.3 Incremental contribution revisited

The paper advances both corporate finance and asset pricing. For corporate finance, it provides an aggregate theory of debt maturity choice that complements firm-level theories such as maturity matching, signaling, and financial constraints. For asset pricing, it shows that nonfinancial firms can be important suppliers of duration and can help absorb preferred-habitat shocks.

The paper also clarifies the interpretation of corporate market timing. If firms issue short-term debt before long-term bonds earn high returns, that does not necessarily mean managers are unusually good forecasters. It may simply mean that firms respond to government supply shocks that are themselves related to expected returns.

8.4 Implications for corporate finance and asset pricing

The findings imply that capital-structure decisions should not be analyzed only at the firm level. Aggregate bond supply, Treasury issuance policy, and limits to arbitrage can influence corporate financing choices. This creates a channel through which public debt management affects private-sector maturity structure.

For asset pricing, the paper suggests that the corporate sector can be part of the arbitrage system. Operating firms may absorb macro risks when specialized arbitrageurs have limited capacity, especially when the relevant adjustment is close to Modigliani-Miller neutral for firms.

8.5 Limitations

The evidence is observational. Although the IV estimates and robustness tests are persuasive, the exclusion restriction for government debt-to-GDP cannot be proven directly. Annual aggregate data also limit the precision with which adjustment dynamics can be identified. Compustat covers public firms and may be affected by compositional changes. Some relevant duration exposures, such as swaps, mortgages, and international debt-market integration, are difficult to measure perfectly. Finally, the main sample ends in 2005, so the paper does not directly study the post-crisis and quantitative-easing period.

8.6 Takeaway

The enduring takeaway is that corporate debt maturity choice is partly a macro liquidity-provision activity. Firms do not need a special forecasting advantage to time the debt market. They can appear to time the market because they absorb maturity supply imbalances created by government debt management. The paper turns corporate debt maturity from a purely firm-level financing decision into a link between Treasury supply, limits to arbitrage, and corporate capital structure.